$$g'(x) = 2x+1$$

$$\frac{df}{dg} = \frac{2x}{(2x+1)(x^4+2x^2+2)}.$$

Example 10.30

Differentiate $\sin(ax^2+bx+c)$ with respect to $\cos(lx^2+mx+n)$

Solution

Let $u = \sin(ax^2+bx+c)$

$$v = \cos(lx^2+mx+n)$$

$$\frac{du}{dv} = \frac{u'(x)}{v'(x)}$$

$$u'(x) = \cos(ax^2+bx+c)(2ax+b)$$

$$v'(x) = -\sin(lx^2+mx+n)(2lx+m)$$

$$\frac{du}{dv} = \frac{u'(x)}{v'(x)} = \frac{(2ax+b)\cos(ax^2+bx+c)}{-(2lx+m)\sin(lx^2+mx+n)}.$$

10.4.8 Higher order Derivatives

If $s = s(t)$ is the position function (displacement) of an object that moves in a straight line, we know that its first derivative has the simple physical interpretation as the velocity $v(t)$ of the object as a function of time :

$$v(t) = s'(t) = \lim_{\Delta t \to 0} \frac{f(t+\Delta t)-f(t)}{\Delta t} = \frac{ds}{dt}.$$

The instantaneous rate of change of velocity with respect to time is called the acceleration $a(t)$ of the object. Then, the acceleration function is the derivative of the velocity function and is therefore the second derivative of the position function:

$$a(t) = v'(t) = \lim_{\Delta t \to 0} \frac{v(t+\Delta t)-v(t)}{\Delta t}$$

$$= \frac{d}{dt}(v(t))$$

$$= \frac{d}{dt}\left(\frac{ds}{dt}\right) = \frac{d^2s}{dt^2} = s''(t)$$

Thus, if f is a differentiable function of x, then its first derivative $f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x)-f(x)}{\Delta x}$ has a very simple geometrical interpretation as the slope of a tangent to the graph of $y = f(x)$. Since f' is also a function of x, f' may have a derivative of its own, and if it exists, denoted by $(f')' = f''$ is,

$$f''(x) = \lim_{\Delta x \to 0} \frac{f'(x+\Delta x)-f'(x)}{\Delta x}.$$

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